

Roll No.:

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech.-CSE)

Semester: 5th

Name of the Course: Computer Networks

Course Code: TCS 511

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1		(20Marks)	
(a)	A server needs to send data to a client using TCP/IP. Explain, with the help of a diagram, the process of encapsulation at different protocol layers.		CO1
(b)	In packet-switched networks, congestion often occurs. Explain with a real-world example, how queuing delay impacts performance in video streaming?		CO1
(c)	Calculate effective throughput if 1000-byte packet sent over 10 Mbps, 50-byte header per packet.		CO1
Q2			
(a)	A University streams online lectures to students. Without CDN, streams buffer frequently. After adopting a CDN, quality improves drastically. Discuss, how CDN improves video delivery in context of edge caching, adaptive bitrate streaming, reduced latency?		CO2 CO2
(b)	A DNS query takes 2 RTTs without caching and with caching it takes only 1 RTT. If RTT = 80 ms. Calculate % reduction in delay.		CO2
(c)	How do cookies work in HTTP? Explain their role in maintaining state between client and server.		
Q3			
(a)	Describe the process of a DNS lookup. How does a client resolve a domain name to an IP address?		CO3 CO3 CO4
(b)	As TCP can be enhanced with TLS to provide process-to-process security services, including encryption. Does TLS operate at the transport layer or the application layer? If the application developer wants TCP to be enhanced with TLS, what does the developer have to do?		
(c)	A network application needs low latency but can tolerate some packet loss (e.g., online gaming). Justify whether UDP, TCP, or QUIC is most appropriate.		
Q4			
(a)	A data file of 20 MB needs to be transferred over a transport layer protocol with 5 Mbps link bandwidth and 50 ms propagation delay. (i) Compute the transmission delay. (ii) Discuss how performance differs if the file transfer uses UDP (without retransmission) vs QUIC (with retransmission + congestion control).		CO4 CO5 CO5
(b)	It has been said that when IPv6 tunnels through IPv4 routers, IPv6 treats the IPv4 tunnels as link-layer protocols. Do you agree with this statement? Why or why not?		
(c)	How does generalized forwarding differ from destination-based forwarding?		
Q5			
(a)	Discuss about the SDN Components? Differentiate SDN with Traditional Networking.		CO5
(b)	Calculate the store-and-forward delay at a single switch for a link rate of $R = 622$ Mbps for $L = 1,500$ bytes, and for $L = 50$ bytes.		CO6
(c)	Why is an ARP query sent within a broadcast frame? Why is an ARP response sent within a frame with a specific destination MAC address?		CO6

